

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Electrical Engineering
2. Subject Name: Introduction to Artificial Intelligence and Machine Learning

L-T-P: 3-0-0

Credits: 3

3. Pre-requisites:
4. Syllabus and reference books:

Syllabus:

Representation with propositional and predicate calculus. State space search structure and strategies. Heuristic search. Knowledge representation. Automated Reasoning. Fuzzy logic. Data compression. Unsupervised, semi-supervised, supervised learning. Probabilistic and distance based decision making. Decision trees and random forest. Neural networks and deep learning. Reinforcement learning.

Reference Books:

- 1) G. F. Luger, Artificial Intelligence, Pearson, 6/e, 2009.
- 2) E. Rich, K. Knight, S. B. Nair, Artificial Intelligence, McGraw-Hill, 3/e, 2009.
- 3) S. Russel, P. Norvig, Artificial Intelligence, Pearson, 2/e, 2003.
- 4) T. H. Cormen, C. E. Leiserson, R. L. Rivest, C. Stein, Introduction to Algorithms, PHI, 3/e, 2009.
- 5) S. Theodoridis, K. Koutroumbas, Pattern Recognition, Elsevier-Academic Press, 4/e, 2009.
- 6) T. M. Mitchell, Machine Learning, McGraw Hill, 1/e, 1997.
- 7) S. Haykin, Neural Networks and Learning Machines, Pearson, 3/e, 2009.
- 8) T. M. Cover, J. A. Thomas, Elements of Information Theory, Wiley, 2/e, 2006.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Artificial Intelligence Representation <i>Propositional calculus, Predicate calculus, Inference rules to produce predicate calculus expressions, Example of building a logic based personal financial advisor, electrical circuit component advisor.</i>	3

2.	Structures and Strategies for State Space Search <i>State-space representation of problems as graph, finite state machines. Example of the Tic-Tac-Toe, Traveling salesperson, 8-puzzle. Forward chaining and data driven search vs. Backward chaining and goal-driven reasoning. Graph search following depth first search and breadth first search. A-star search. State space to represent reasoning with propositional and predicate calculus. Example of solving mathematical equations, electrical circuits, and the logic based financial advisor.</i>	6
3.	Heuristic Search <i>Hill climbing, Dynamic programming. Best first search. Admissibility, Monotonicity and Informedness. Genetic algorithms.</i>	3
4.	Knowledge Representation <i>Conceptual graphs. Alternative Representations and Ontologies. Agent based and distributed problem solving. Example of process modelling viz. cooking, surgery, etc.</i>	3
5.	Automated Reasoning <i>Weak methods of theorem proving. General problem solver and difference tables. Resolution theorem proving. Automated reasoning. Application to power demand estimation.</i>	3
6.	Fuzzy Logic and Controllers <i>Fuzzy sets and fuzzy number system. Fuzzification, Fuzzy Rules, Inference engine, Defuzzification. Building Fuzzy logic controller for electrical appliances.</i>	3
7.	Data Compression with Unsupervised Learning <i>Lossy compression system. Principal component analysis. Singular value decomposition. Lossless compression system. Huffman tree and the Huffman encoding.</i>	3
8.	Probabilistic Decision Making with Supervised Learning <i>Concept of histogram, bins, probability mass. Bayes' classification rule. Maximum a posteriori (MAP) and maximum likelihood estimation (MLE). Application to electrical apparatus and electrical power flow control.</i>	3

9.	Distance based Decision Making and Semi-supervised Learning <i>Nearest neighbor search. K-means clustering. Fuzzy c-means clustering. Application to pattern discovery and decision making.</i>	3
10.	Decision Trees and Random Forest <i>Binary decision trees. Split criteria. Information gain. Decision tree aggregation rules. Random forest inference. Application to signal and image classification.</i>	3
11.	Neural Networks and Deep Learning <i>Feedforward neural networks. Convolution neurons. Activation functions. Loss functions. Training of neural networks. Autoencoders. Application to handwritten digit and image classification. Recurrent neural networks. Application to time series data analysis, signal classification, electrical power demand estimation.</i>	3
12.	Reinforcement Learning <i>Policy evaluation and search. Reward function. Value function. Model of the environment. Q-learning, policy gradients, temporal-difference learning. Application to electrical power system stability, training of neural networks, machine learning architecture design.</i>	3
Total number of hours		39